

Yuvraj Kumar Jaiswal

📍 Raigarh, Chhattisgarh, India ✉ yuvrajkumar92081@gmail.com 📞 +91 97540 70225

🌐 in/yuvraj-kumar-jaiswal-832559252 🐙 github.com/Yuvrajkumar46

SUMMARY

AI-focused Software Engineer with hands-on experience in building RAG-based (Retrieval-Augmented Generation) systems using FastAPI, FAISS, and LLMs. Skilled in designing end-to-end AI pipelines, semantic search, and context-aware applications. Strong foundation in DSA, backend development, and scalable API design with a focus on AI-native engineering.

PROJECTS

RAG-Based AI Document Q&A System 🔄

Tools: Python, FastAPI, FAISS, LLM (Llama 3 via Groq), REST APIs

- Built an end-to-end **Retrieval-Augmented Generation (RAG)** pipeline for document-based question answering.
- Implemented **document parsing, chunking, and embedding generation**, storing vectors using **FAISS** for efficient semantic search.
- Integrated **LLM for context-aware response generation**, reducing hallucination using retrieval grounding.
- Designed and exposed **REST APIs using FastAPI** for document upload and query handling.
- Supported **PDF/TXT ingestion** with real-time query processing and low-latency retrieval.
- Developed modular architecture separating **embedding, retrieval, and LLM services**.
- Improved response accuracy through optimized chunking and retrieval strategies.

GreenGraph – Project Allocation System 🔄

Tools: HTML, CSS, JavaScript, MySQL, Git, GitHub

- Developed a web-based system to automate academic project allocation using **logic-driven algorithms**.
- Implemented **role-based authentication** (Admin, Student, Faculty) and allocation based on **average CPI**.
- Designed normalized **MySQL database schemas** and optimized SQL queries for data consistency.
- Followed structured **SDLC practices** ensuring maintainability and reliability of the system.

Nyaya – AI-Based Legal Advisory Platform 🔄

Tools: React.js, Node.js, Express.js, MySQL, REST APIs, Socket.IO, JWT, AI Integration

- Built a full-stack **AI-powered legal advisory platform** enabling users to get instant legal guidance using **LLM-based query processing**.
- Designed scalable **backend services** using **Node.js and Express** with modular architecture and clean REST API design.
- Implemented **RBAC** and secure authentication using **JWT and bcrypt**, ensuring protected access across multiple user roles.
- Integrated **real-time communication** using **Socket.IO** for live consultations and interactive user experience.
- Developed and optimized **MySQL database schemas** and built a document management system with upload and sharing features.

Mini Projects

- Built multiple **Python-based backend applications** and REST-style services to strengthen system design and DSA.
- Developed frontend interfaces using **HTML, CSS, JavaScript** for rapid prototyping.
- Explored **AI data pipelines and preprocessing workflows** using Python.
- Gained hands-on experience in **debugging, testing, and performance optimization**.

EDUCATION

Bachelor of Technology in Computer Science	2022–2026
Vellore Institute of Technology	CGPA: 8.16
CBSE (XII)	2021–2022
O.P. Jindal School, Raigarh, Chhattisgarh	Score: 88.6%
CBSE (X)	2019–2020
St. Michael's Sr. Sec. School, Raigarh, Chhattisgarh	Score: 90.6%

SKILLS

Programming Languages: Python, SQL

Web & API Development: FastAPI, REST APIs, HTML, CSS

AI & LLM: Retrieval-Augmented Generation (RAG), LLM Integration (Llama 3), Prompt Engineering, Embeddings, Semantic Search, FAISS

Libraries & Tools: NumPy, Pandas

Databases: MySQL

Tools & Platforms: Git, GitHub, Linux (Basics), VS Code, AWS (Basics)

Backend & System Design: API Architecture, Modular Design, Object-Oriented Programming, Scalable Systems

Professional Skills: Technical communication, stakeholder collaboration, documentation

Core Concepts: Data Structures & Algorithms, DBMS, Operating Systems, Computer Networks, Machine Learning Fundamentals

CERTIFICATIONS

Data Science Professional – Oracle | Data Science with Python – IAMNEO
Python Essentials – Vityarthi | The Bits and Bytes of Computer Networking – Coursera